

Adi Karthikeya Sunkara Balasubramanyam

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SUMMARY

Data Scientist and ML Engineer with hands-on experience building end-to-end machine learning pipelines, deploying containerized AI services, and delivering data-driven insights through statistical modeling and dashboards. MS in Data Science (GPA 3.89, University of Houston, May 2026) with expertise in Python, SQL, NLP, deep learning, LLMs, and cloud deployment (GCP, AWS, Docker). Ranked 4th globally at the ODSC AI & Google Cloud Agentic AI Hackathon. Available on OPT (STEM extension eligible).

EDUCATION

University of Houston <i>Master of Science in Data Science; GPA: 3.89 / 4.00</i> Amrita Vishwa Vidyapeetham <i>Bachelor of Technology in Electronics & Communication Engineering; GPA: 3.50 / 4.00</i>	Houston, TX Aug 2024 – May 2026 Bengaluru, India Aug 2020 – Jun 2024
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EXPERIENCE

Data Science Intern <i>NSF Data Science for Energy Transition (Federally Funded)</i> Engineered ML pipelines using PCA, Random Forest, and XGBoost for geological facies classification on 50K+ samples, achieving 80.24% accuracy and a 7% improvement over baseline logistic regression. Built interactive Tableau dashboards and QGIS visualizations synthesizing multi-source subsurface data into actionable insights, directly informing energy research investment decisions for NSF stakeholders. Automated data preprocessing and feature extraction using pandas/NumPy, reducing pipeline execution time by 35%.	May 2025 – Jun 2025 Houston, TX
Graduate Teaching Assistant <i>University of Houston</i> Designed and graded ML/data analytics assignments for 120+ graduate students, enforcing reproducibility standards across Python-based pipelines using scikit-learn, pandas, and Jupyter notebooks. Developed automated grading scripts in Python to evaluate model accuracy and code quality, reducing manual evaluation time by 40% and standardizing feedback across 15+ weekly submissions. Mentored students on end-to-end ML workflows including data preprocessing, feature engineering, model selection, and hyperparameter tuning using GridSearchCV and cross-validation.	Jan 2025 – Present Houston, TX
Data Analyst <i>Corizo Edutech</i> Automated ETL and feature engineering pipelines using Python (pandas, NumPy, scikit-learn) across 15,000+ records, reducing manual preprocessing time by 60% and improving downstream data quality for reporting. Built classification and regression models (Random Forest, XGBoost) with GridSearchCV tuning, boosting prediction accuracy by 12%; delivered weekly Tableau dashboards tracking 8 KPIs across 3 business units. Wrote complex SQL queries (MySQL) to extract, aggregate, and join multi-table datasets for A/B testing and cross-functional reporting across product and operations teams.	Apr 2023 – May 2024 Bengaluru, India

PROJECTS

ShopperAI – Agentic AI Shopping Assistant <i>Python, Google ADK, Gemini, GCP, Docker</i> Ranked 4th globally out of ~200 teams at the ODSC AI & Google Cloud Agentic AI Hackathon; built production-grade agentic AI with deterministic, stage-based workflows for product discovery and virtual try-on, deployed on GCP Cloud Run. Engineered modular agent orchestration with memory management, reducing hallucinated outputs by 30% and improving recommendation relevance by 25%. Integrated Gemini API with GCP-hosted backend, enabling real-time personalized recommendations across 10K+ product catalog.
Real-Time Customer Churn Prediction Pipeline <i>Python, SQL, XGBoost, MLflow, Docker, Streamlit</i> Built end-to-end ML pipeline (XGBoost + LightGBM) achieving ROC-AUC 0.87 and 83.2% accuracy on 50K+ records; wrote SQL queries for feature extraction including CLV and RFM segmentation. Deployed as Streamlit web app with Docker containerization and MLflow experiment tracking, enabling <200ms real-time inference; implemented SHAP-based explainability dashboards for business stakeholders.
RAG-Powered Document Intelligence System <i>Python, LangChain, ChromaDB, FastAPI, Docker</i> Built production-ready RAG pipeline processing 1,000+ documents with 92% retrieval accuracy; FastAPI backend with async endpoints achieving <500ms response time and 99.5% uptime. Implemented RAGAS evaluation pipeline (faithfulness, relevance, context precision), improving retrieval quality by 18%.

TECHNICAL SKILLS

Languages: Python, SQL, R
ML & AI: scikit-learn, PyTorch, TensorFlow, HuggingFace Transformers, LangChain, XGBoost, LightGBM
NLP & LLMs: BERT, GPT, SBERT, RAG, Prompt Engineering, Fine-tuning, RLHF
Data & Visualization: pandas, NumPy, Tableau, Power BI, Matplotlib, Seaborn, Microsoft Excel
Cloud & MLOps: GCP (Cloud Run, BigQuery, Vertex AI), AWS (S3, EC2), Docker, MLflow, FastAPI, Git/GitHub, CI/CD
Databases & Web: PostgreSQL, MySQL, MongoDB, Flask, FastAPI, Streamlit, Django